# Swati Aggrawal

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## PROFESSIONAL SUMMARY

MSc Computer Science student at Sapienza Università di Roma specializing in Machine Learning and Artificial Intelligence. Experienced in developing machine learning and NLP solutions, evaluating model performance, building automated AI workflows, and deploying applications using Python, FastAPI, Docker, and AWS. Strong background in Generative AI, transformer models, embeddings, prompt engineering concepts, machine learning experimentation, and model evaluation. Published IEEE researcher with hands-on experience translating business requirements into practical AI-driven solutions. Seeking AI/ML Engineer and Applied AI roles.

## EDUCATION

**Sapienza Università di Roma** **May 2026**

*Master of Science in Computer Science*

- **Relevant Coursework:** Deep Learning and Applied AI, NLP, Computer Vision, Distributed Systems.
- **Thesis:** Evaluating Pretrained Vision Transformers Under Continual Learning.
- Selected for a competitive international master's program.

**Sharda University** **July 2023**

*Bachelor of Technology - CSE with AI and ML*

- **Relevant Coursework:** Operating Systems, Data Structures, Machine Learning, Databases, Cloud Computing
- **Thesis:** Virtual Generative Model Inducing Human Identity to Computers in Metaverse
- Awarded **merit-based scholarship** (Top 5–10% of cohort).

## TECHNICAL SKILLS

- **Programming :** Python, Bash, SQL, C++
- **Machine Learning :** Machine Learning, Classification, Regression, Feature Engineering, Model Evaluation, Cross Validation, Hyperparameter Tuning, Statistical Analysis, Data Preprocessing
- **Generative AI & NLP :** Large Language Models (LLMs), Transformer Models, Prompt Engineering, Few-Shot Prompting, Embeddings, Sentence Transformers, Semantic Similarity
- **Libraries & Frameworks :** Pandas, NumPy, Scikit-Learn, PyTorch, TensorFlow, Keras, Sentence Transformers
- **Deployment & Cloud :** FastAPI, REST APIs, Docker, AWS ECS, AWS Fargate, CloudWatch
- **Tools :** Git, GitHub, Linux

## PROJECTS

### DocDelta [Document Comparison API]

- Developed an NLP-powered document intelligence system using transformer embeddings and semantic similarity techniques for automated comparison of legal and contract documents.
- Designed algorithms to identify added, removed, and modified content while distinguishing numerical changes from textual modifications.
- Built an automated AI workflow that reduced manual document review effort and enabled scalable processing of document revisions.
- Leveraged Sentence Transformers and embedding-based retrieval techniques to analyze semantic similarity between document versions.
- Deployed the application using FastAPI, Docker, AWS ECS Fargate, and CloudWatch for scalable inference and monitoring.
- Evaluated inference performance and optimized deployment architecture, achieving approximately 840ms warm inference latency.
- Experimented with prompt engineering techniques and transformer-based language models to improve semantic analysis workflows.

### Lossy Text Compressor [Semantic NLP Pipeline]

- Developed an embedding-based NLP pipeline using GTR-T5 transformer models, vector quantization, and text reconstruction to analyze semantic compression-quality trade-offs.

- Performed systematic experiments across multiple quantization levels and evaluated results using cosine similarity and ROUGE-L metrics.
- Achieved optimal performance at 6-bit quantization with cosine similarity of 0.964 and ROUGE-L score of 0.776.
- Identified non-linear relationships between compression levels and reconstruction quality, providing insights for efficient semantic storage and retrieval.
- Conducted quantitative analysis and model evaluation to determine optimal compression-performance trade-offs.

## RESEARCH EXPERIENCE

### Continual Learning for Computer Vision (Master's Thesis) — Nov 2025

- Designed and executed machine learning experiments to evaluate Vision Transformer performance across sequential learning tasks.
- Designed reproducible machine learning experimentation pipelines and comparative evaluation frameworks for transformer-based models.
- Analyzed model performance using quantitative evaluation metrics to measure knowledge retention and catastrophic forgetting.
- Investigated catastrophic forgetting and knowledge retention in transformer-based visual learning systems. .
- Automated experiment tracking and performance benchmarking across multiple continual learning scenarios.
- Documented experimental findings and identified research opportunities for future development of continual learning systems.
- **Technologies :** Python, PyTorch, Vision Transformers, Computer Vision

### Machine Learning for Air Pollution [IEEE Publication] — May 2023

- Published an IEEE conference paper reviewing machine learning approaches for air quality prediction.
- Analyzed six machine learning models including Random Forest, SVM, Genetic Algorithms, LS-SVM, Decision Trees, and Artificial Neural Networks.
- Evaluated prediction-based and knowledge-discovery methodologies across six NAAQS pollutants.
- Identified real-time monitoring and IoT integration as important future directions for environmental analytics.